AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM352
# Web, Mobile and Cloud Technologies

## Midterm Examination (MTA)-Makeup
## Spring Semester 2024/2025

## Date: 23/04/2025

| | |
|---|---|
| **Number of Exam Pages: 4** (including this cover sheet) | **Time Allowed: 2 Hrs** |

**Instructions:**

- Total Marks: 45 {will be adjusted to 30 marks}
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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# PART I (5 Marks)

## Answer all the questions

**Choose the right answer**

1. HTTP is defined as stateless protocol, because it makes no information about
   a. Client
   b. Server
   c. Message Type
   d. Network Configuration
2. JSON uses _____ to represent an array
   a. Pair of square brackets []
   b. Pair of angular brackets<>
   c. Pair of flower brackets{}
   d. Pair of brackets()
3. Which attribute of the form element tells how the data is transmitted.
   a. Method
   b. Action
   c. Enctype
   d. name
4. "A client cannot access the online shop of a vendor's company due to an intensive number of fake shopping requests". Which class of security threat this case falls into.
   a. Fabrication
   b. Modification
   c. Interruption
   d. Interception
5. REST web services are described as stateless, meaning that
   a. Some information can be stored on server
   b. No information can be stored on server
   c. Rest has no operations
   d. Rest can not be a web service

# PART II (20 Marks)

## Answer all the questions

1. Discuss Symmetric Key encryption. Give one advantage and one disadvantage of Symmetric Key encryption. [4 Marks]
2. Discuss GET, POST, PUT, HEAD methods of HTTP. [6 marks]
3. Discuss four categories of attackers from which distributed systems need to be protected. [4 marks]
4. Provide XML representation for given JSON data [6 marks]

```
[
 {
  "ID": 1,
  "Name": "John",
  "Age": 28,
  "Email": "john@example.com"
 },
 {
  "ID": 2,
  "Name": "Alice",
  "Age": 24,
  "Email": "alice@example.com"
 },
 {
```

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```
  "ID": 3,
  "Name": "Bob",
  "Age": 32,
  "Email": "bob@example.com"
 }
]
```

# PART III (20 Marks)

## Answer all the questions

1. Given the SOAP web service "Calculator". Complete the web service operation named "meter2KM" that converts the entered meter value to Kilo Meter by dividing the meter value by 1000 and return the KM value. [6 Marks]

```
@_____(1)_____(serviceName = "calculator")
public class calculator {
/**
   * Web service operation
**/
@_____(2)_____(operationName = "meter2KM")
 public String __(3)_____(@_____(4)_____(name = "meter") double __(5)__) {

    ___(6)_____ ""+Math.round((meter/1000)*100.0)/100.0;
  }
}
```

2. Complete the missing xml elements from the below SOAP Response messages. [5 marks]

*The following are the SOAP Request and Response messages for the above soap service given in Question 1.SOAP Request*

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope
xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:Header/>
    <S:Body>
        <ns2:meter2KM xmlns:ns2="http://calci/">
            <meter>1500.0</meter>
        </ns2:meter2KM>
    </S:Body>
</S:Envelope>
```

*SOAP Response*

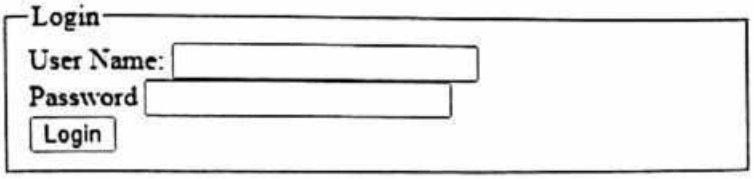
```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope
xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:_____/>
    <S:Body>
        <ns2:_____ xmlns:ns2="http://calci/">
            <_____>
        </ns2:_____>
    </S:Body>
</S:Envelope>
```

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3. Design a simple registration form using HTML5 with input fields as given in figure-1 below. Use proper input type for each field. [6 Marks]




2. Modify the following HTTP response to express that the page requested is not found and the request date is today [3 marks]

```
HTTP/1.1 200 OK
Date: Mon, 6 November 2015
20:52:00 GMT
Server: Apache
Last-Modified: Mon, 21 Dec 2015
11:17:55 GMT
```

**End of questions**

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